

statsmodels.stats.stattools.jarque_bera

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statsmodels.stats.stattools.jarque_bera(resids, axis=0) \[source\]
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The Jarque-Bera test of normality.

Parameters

resids : *array_like*

Data to test for normality. Usually regression model residuals that are mean 0.

axis : *int, optional*

Axis to use if data has more than 1 dimension. Default is 0.

Returns

- **JB** (*{float, ndarray}*) – The Jarque-Bera test statistic.
- **JBpv** (*{float, ndarray}*) – The pvalue of the test statistic.
- **skew** (*{float, ndarray}*) – Estimated skewness of the data.
- **kurtosis** (*{float, ndarray}*) – Estimated kurtosis of the data.

Notes

Each output returned has 1 dimension fewer than data

The Jarque-Bera test statistic tests the null that the data is normally distributed against an alternative that the data follow some other distribution. The test statistic is based on two moments of the data, the skewness, and the kurtosis, and has an asymptotic χ^2_2 distribution.

The test statistic is defined

$$JB = n(S^2/6 + (K - 3)^2/24)$$

where n is the number of data points, S is the sample skewness, and K is the sample kurtosis of the data.

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